Ibtihel JALEL

Embedded Software Validation Engineer | STM32 | C/C++ | Python

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Profile

Embedded Software Validation Engineer with **1 year of professional experience at STMicroelectronics**, specializing in **STM32 firmware validation, test automation and software integration**. Experienced in developing **Python/Bash automation tools, validation workflows and embedded test use cases**, with hands-on expertise in **C/C++, FreeRTOS, functional, integration and regression testing, failure analysis and HW/SW integration**. Familiar with **Git, Gerrit, Jenkins and Jira** in collaborative CI and validation environments.

Technical Skills

Programming: C, C++, Python, Bash
Embedded: STM32, FreeRTOS, CMSIS, HAL/LL, firmware development, HW/SW integration
Test & Validation: Functional, integration, regression, robustness testing, test campaigns, failure analysis, root-cause analysis
Automation & Scripting: Test automation, validation frameworks, GUI development, project/build automation
Protocols: I2C, UART, SPI, USART
Development & CI: Git, Gerrit, Jenkins, CI/CD, Jira, Bugzilla, Linux
MCU Tools: IAR Embedded Workbench, STM32CubeMX, STM32CubeIDE, STM32CubeProgrammer, Keil μVision, VS Code
Laboratory: Oscilloscope, signal generator, voltage/current measurements
AI / ML: Machine Learning, Random Forest, LSTM

Professional Experience

Embedded Software & Validation Engineer – STMicroelectronics, Ariana, Tunisia *Oct. 2025 – Sep. 2026*

- Executed **functional, integration, regression and robustness validation campaigns** for STM32 peripherals and embedded software; analyzed and reproduced failures, performed root-cause analysis, and raised and tracked Jira bugs.
- Developed a script to **automatically enable STM32 peripherals, generate IAR projects and compile firmware** using STM32CubeMX2, accelerating project setup and validation.
- Developed scripts to **test MX2 performance** and automate performance-oriented validation activities.
- Used **GitHub Copilot** with structured prompts to develop an automation script that extracts test results from Jenkins reports and updates Jira test executions, improving validation traceability and reducing manual effort.
- Developed a **dual-STM32C5 FreeRTOS use case** with concurrent sensor acquisition, data processing, security and real-time monitoring; integrated temperature and gyroscope acquisition, AES encryption/decryption, CRC integrity checks, I2C/USART communication, ADC/DAC and VCP/USART monitoring.
- Worked with **Git, Gerrit and Jenkins** in CI workflows, contributing to validation automation and tooling.

Embedded Software Intern – STMicroelectronics, Ariana, Tunisia *Feb. – Aug. 2024*

- Developed an **automated STM32 driver validation framework** using a dedicated test board communicating with the DUT through **I2C**.
- Automated firmware adaptation and validation workflows across **STM32L5, STM32C0 and STM32F4** platforms.
- Developed a **Python GUI and Random Forest model** to assist test-board selection according to device requirements.
- Implemented and validated **RCC clock calibration** test cases.

Embedded Software Intern – STMicroelectronics, Ariana, Tunisia *Jul. – Sep. 2023*

- Maintained and updated the **STM32U0 firmware validation environment** and ensured compatibility with updated CMSIS components.
- Developed and executed automated and manual test cases for **Stop and Standby low-power modes**; analyzed results and maintained technical documentation.

Electrical Engineer Intern – YAZAKI, Gafsa, Tunisia *Jul. – Aug. 2022*

- Supported quality-control activities and technical training related to **automotive electrical systems**.

Education

National Engineering School of Monastir (ENIM) – Engineering Degree in Electrical Engineering, Industrial Computing and Instrumentation *2021 – 2024*

Faculty of Sciences of Sfax – Preparatory Cycle for Engineering Studies, Physics and Chemistry *2018 – 2021*

Projects

Smart Greenhouse System – Embedded / IoT
Developed a real-time monitoring and control system using **STM32F4, ESP32, Embedded C, FreeRTOS, Firebase and Flutter**; implemented sensor acquisition, inter-board communication and embedded data processing.

Lithium-Ion Battery State-of-Charge Prediction – Machine Learning
Developed an **LSTM-based model** for battery State-of-Charge prediction, achieving approximately **95% prediction accuracy**. *Python, LSTM, Machine Learning*

Bank Payroll Management System – C++ / Qt
Developed a desktop payroll management application using **C++ and Qt**.

Certifications & Languages

Certifications: Cortex-M Software Development Fundamentals – ARM | Scrum Fundamentals – SCRUMstudy | Python for Data Science, AI & Development – IBM/Coursera
Languages: Arabic – Native | French – C1 | English – B1